Dresden University of Technology

Computational Logic ∴ Group

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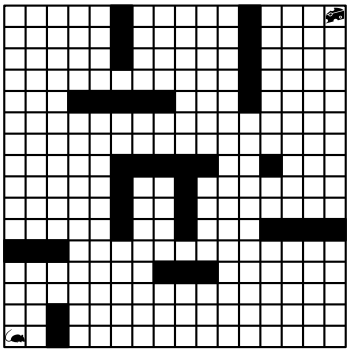
Advanced Problem Solving and Search

Winter Term 2026/27

Exercises 2

5/11/2026–9/11/2026

Problem 1.
Design a Simulated Annealing Algorithm for the Maze Problem below. The goal is to find the shortest path in the maze, starting from the mouse and ending in the cheese.



Consider the following questions.

a) How would do you represent a solution? If you are considering different options, discuss the pros and cons.

b) How do you generate the initial solution resp. how do you ensure feasibility of solutions?

c) What is the neighbourhood of a solution?

d) What is the evaluation function?

e) Compare the SA approach to solving the maze problem with A*. For each algorithm, state whether it is *complete* and *optimal* for this problem, and briefly argue which algorithm you would prefer here and why.

Problem 2.
Consider the following propositional formula in conjunctive normal form (CNF) over variables x_1, x_2, x_3:

$$\varphi = \underbrace{(x_1 \vee x_2)}_{C_1} \wedge \underbrace{(\neg x_1 \vee x_3)}_{C_2} \wedge \underbrace{(\neg x_2 \vee \neg x_3)}_{C_3} \wedge \underbrace{(x_1 \vee \neg x_2 \vee x_3)}_{C_4}$$

1. **GSAT trace.** Starting from the truth assignment $A_0 = (x_1 = \mathbf{F}, x_2 = \mathbf{T}, x_3 = \mathbf{F})$, execute **three steps** of GSAT. At each step, list the number of unsatisfied clauses for every possible single-variable flip, identify the flip chosen by GSAT, and state the resulting assignment.
2. **Local optimum.** After the three steps in (a), determine whether the current assignment is a local optimum under the GSAT neighbourhood (i.e., no single flip reduces the number of unsatisfied clauses). Justify your answer.
3. **Stochastic acceptance.** Suppose the current assignment is a local optimum with $eval(v_c) = 107$ (number of satisfied clauses in a larger formula instance), and the only available flip yields a neighbour with $eval(v_n) = 104$. Using the stochastic hill-climber acceptance formula with temperature $T = 5$, compute the acceptance probability p and decide whether the move is accepted given a random draw of $r = 0.61$.
4. **Role of temperature.** Using the same $\Delta = eval(v_c) - eval(v_n) = 3$, compute p for $T \in \{1, 10, 100\}$. What do you observe about the behaviour of the algorithm as T increases?